

Ethical Implications of Addictive Business Models: A Comparative Analysis of Social Media and other Products

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Abstract— This paper examines the ethical implications of business models that incentivize addiction, focusing on social media, gambling, and video games. It argues that products designed primarily to keep users engaged indefinitely are unethical because they exploit users for profit and can cause significant harm. By comparing these models with traditional entertainment and assessing them through Kantian ethics and rule utilitarianism, the paper highlights the need for regulatory measures and educational strategies to mitigate the negative impacts of addictive design while acknowledging the benefits these platforms provide

Keywords— *Addictive Design, Ethics of Business Models, User Engagement, Social Media Regulation,*

I. INTRODUCTION

“Addiction is a treatable, chronic medical disease involving complex interactions among brain circuits, genetics, the environment, and an individual’s life experiences. People with addiction use substances or engage in behaviors that become compulsive and often continue despite harmful consequences” [1].

Addiction is a very serious problem in many parts of the world. This is why social media business models that promote addiction are so worrisome. Social media’s current model is to maximize engagement [2]. They do this because they can sell the data gathered from their users to advertisers while the users are engaging with the platform. This is concerning because it incentivizes social media sites to implement addictive designs to keep their users engaged and online for as long as possible. A business that is purely financially incentivized to addict its users is unethical and should be regulated, especially when the products or business’ only income is derived from the customer’s or user’s continuous engagement with the product.

II. WHAT PRODUCTS ARE A PROBLEM

Problematic addiction can arise from almost anything, but a typical business model is not purely incentivized to make its product addictive. A normal business model involves creating a product that people want to buy; people buy the product, and that completes the transaction. This business model incentivizes the creation of products that customers will want to buy again. While the product could be addictive, the seller’s

primary concern is not how much time the customer spends with the product but rather that the product is purchased.

All forms of entertainment can be addictive because people naturally want to experience enjoyable activities more frequently. However, a problem arises when entertainment is designed to be addictive rather than simply as entertaining as possible. If a product’s goal is to provide entertainment, it should focus on engaging users so they return because they genuinely enjoy it. The goal should not be to get users addicted to the point where they spend all their time on the product. Instead, the aim should be for users to spend their free time and excess money on the product or service because they find it genuinely enjoyable and worthwhile.

A. Gambling

Gambling companies are notorious for trying to keep their customers gambling for as long as possible. Las Vegas casinos are designed so that customers lose track of time, encouraging them to keep playing. They are financially incentivized to do this because, on average, the longer a customer plays on a slot machine, the more money the casino gains. This incentive naturally leads to addictive design. Fortunately for casinos, they don’t have to try very hard to make their product addictive. Since people are generally bad at calculating probability, gambling is inherently addictive. Casinos enhance this by not having clocks visible, making it easy to lose track of time, and by creating an environment filled with sounds of people winning, which makes it seem like you could win at any moment. Additionally, they provide everything needed to keep gambling, including food, drinks, and money. Gambling is heavily regulated in most countries, including Canada, precisely because it is so addictive. The fact that gambling is incentivized to be addictive and that casinos implement addictive designs makes it similar to social media.

B. Netflix

Netflix is a close example of a product that uses user data. Unfortunately, Netflix has its own problems with addictive design; the auto-play feature, which incentivizes users to continue watching, is the most notable. However, Netflix is not primarily financially incentivized to make its product addictive. Netflix’s largest incentive is for users to keep their monthly

subscription, which is different from encouraging continuous, excessive use. Netflix would be content if users watched something only once a month, as long as they maintained their subscription.

Netflix uses user data to curate personalized content suggestions. Additionally, Netflix has recently started an ad-supported subscription plan, and its terms of service leave it open to selling user data to advertisers for targeted ads [3]. Nevertheless, Netflix's revenue is not primarily supported through user data, making it less incentivized to use addictive design, which differentiates it from social media platforms.

C. Video Games

Video games are often accused of being addictive, and they certainly can be. Depending on the monetization structure a video game has, developers may also be incentivized to create addictive designs. Since video games are a form of entertainment, they are incentivized to be as entertaining as possible to keep users engaged.

The classic monetization method of buying a video game at full price up front provides no incentive for developers to create addictive designs. The financial incentive can unfortunately lead to over-promising and under-delivering because the developer's primary goal is to get the customer to buy the game once.

A subscription-based model leads to a scenario similar to Netflix; developers want users to keep paying the subscription. The downside is that developers are incentivized to make a game longer to keep players playing and paying the subscription. Luckily, this incentive can clash with the incentive to make an entertaining product. If developers draw out a game too much, it will become less entertaining.

The last common monetization method is microtransactions. Many games add microtransactions on top of other forms of monetization, but some games rely solely on microtransactions. Those that do are incentivized to make the purchases as attractive as possible, as it is the game's only source of revenue. This can play out in multiple ways. Some microtransactions give progression within a game, which is not particularly aligned with addictive design. Unfortunately, some microtransactions are very similar to gambling, where a player buys a random in-game reward which can have differing values. This similarity to gambling comes with the same downsides of addiction. So even if the game itself doesn't implement addictive design, the microtransactions are inherently addictive. If a game relies solely or mostly on this form of monetization, then it shares the same problems as social media.

D. Television

Television also generates significant revenue from advertisements. Luckily for television, most providers are not solely incentivized by ads, as they often have a subscription cost as well. This is a key difference; similar to other products

with a subscription model, they are incentivized to keep users paying for the subscription. Unfortunately, because television also earns a lot of money from advertising, there is a risk of being incentivized to use addictive designs. The addictive nature of television was initially a concern for many people. [4].

E. Professional Sports

Very related to television is professional sports. Professional sports are packed with advertisements. Luckily, watching professional sports is either through television, which is not incredibly addictive, or in person. If you watch sports in person, you need to buy tickets, incentivizing event organizers to sell tickets rather than keep you at the stadium indefinitely. Additionally, athletes would grow tired if they were constantly playing, which would make the entertainment of the sports less enjoyable.

F. Google

A good counterargument to the protective measures is that Google makes most of its money from selling user data. Luckily, Google already has protective measures built in. When you search for something using Google, your goal is to find your answer as quickly as possible. To attract and keep users, Google's largest incentive is to provide the requested information as swiftly and accurately as possible. Therefore, even though Google is incentivized to keep users on the platform to harvest their data, doing so excessively would likely be at the expense of the quality of their product. These competing incentives help ensure that Google does not create an addictive product.

III. FREE WILL VS PROTECTION

If you want to keep someone safe, you might think of wrapping them in bubble wrap and confining them to their house. However, you can't protect someone from every potential danger; there is always a small chance of unforeseen events, like a big storm knocking over a tree and crushing the house. Moreover, keeping someone confined, even in their own home, is generally considered unacceptable and contrary to principles of personal freedom.

In society, we generally avoid restricting each other's basic rights. The UN outlines a set of fundamental rights that everyone should have, including the right to freedom, which implies that individuals should be able to do as they please, provided they do not harm others, as this would infringe upon others' rights [5]. The question arises: how much potential for self-harm should there be before something should be restricted?

For example, while many people believe that suicide should not be permitted, some countries do allow it under certain circumstances [6]. Conversely, activities like skydiving, despite their higher-than-normal risk of harm, are generally not restricted. This reflects a complex balance between respecting personal freedoms and managing risks to oneself.

The line between personal freedom and protection from harm becomes even more complex when considering substances like drugs and alcohol. Different countries approach these issues in varying ways. For instance, some countries have outright prohibited alcohol consumption, while others have not. In the USA and Canada, alcohol was banned at certain points in history, illustrating a shifting stance on managing potential harms associated with its use[7,8].

This historical back and forth reflects broader discussions about whether substances and activities should be restricted. Questions arise about the legality of hard drugs, tobacco, gambling, and marijuana. Should hard drugs be illegal to use? Should tobacco be banned? Is it necessary to prohibit gambling? Should marijuana remain illegal, or is there a case for its decriminalization or legalization? These debates center on finding a balance between respecting personal freedoms and implementing measures to protect public health and safety.

The debate between personal freedom and protection becomes more nuanced when considering minors. In Canada, the answer to whether certain substances and activities should be illegal is generally affirmative for minors. This is largely due to the fact that minors have not yet fully developed their frontal lobes, which are crucial for decision-making and impulse control [9]. As a result, substances such as drugs, tobacco, alcohol, and marijuana are often prohibited for children. These substances are chemically addictive, and the craving for their effects can lead to addiction. Allowing minors access to these addictive products before they have the maturity to fully understand the potential consequences is widely regarded as problematic. This concern is a significant reason behind the restriction of these substances for minors. Similarly, gambling is also prohibited for minors in Canada due to its potential to lead to addiction and other adverse consequences.

In Canada, the legal framework acknowledges that adults are permitted to engage in activities that could potentially harm themselves, provided these activities do not endanger others. For instance, while individuals are allowed to consume alcohol, laws prohibit driving under the influence to prevent putting others at risk. Similarly, smoking tobacco is permitted, but regulations are in place to protect non-smokers from the dangers of secondhand smoke.

However, gambling presents an anomaly in this context. While it is legal, and there are regulations to prevent underage gambling, there are no laws preventing adults from gambling away all their money, even if it results in significant harm to themselves or their dependents. This gap highlights a discrepancy where, unlike other harmful activities, gambling does not face the same restrictive measures despite its potential to cause considerable personal and social harm.

Gambling and social media differ from chemically addictive drugs in that the sellers of these products have distinct incentives. For drugs, the primary goal is often to sell the

product, with less concern for the ongoing use or the well-being of the user once the sale is complete. In contrast, social media platforms are specifically designed to maximize user engagement, which directly influences their revenue through the sale of user data to advertisers. Gambling shares similarities with social media in this regard; it thrives on prolonged engagement, but the financial model often focuses on keeping users gambling rather than on the product itself.

Moreover, the experiences of addiction in both gambling and social media share common characteristics. The immersive, trance-like state experienced by gamblers while engaging with their games is quite similar to the "zone" that users enter when scrolling through social media [10]. This parallel suggests that both gambling and social media can create similarly compelling and potentially addictive experiences for users.

Social media undoubtedly poses risks of addiction, raising the question of whether it should be regulated or even banned. The debate becomes particularly intense when considering the possibility of prohibiting social media use for minors. While adults generally have the freedom to engage in activities that may harm them, minors are often protected from such risks due to their developmental stage and limited decision-making abilities. Unlike substances like drugs and gambling, social media offers significant benefits, such as facilitating connections and communication, which complicates the argument for an outright ban.

Banning social media for minors could be an extreme measure, and enforcing such a ban would likely be impractical. Instead, a more balanced approach might be to focus on education. Just as schools implement mandatory education programs for drugs and alcohol, introducing comprehensive education about social media could help minors understand the risks of addictive design and manage their usage responsibly. This educational approach would aim to mitigate the potential harms of social media while acknowledging its positive aspects

IV. ETHICAL ANALYSIS

A. *Kantiansim*

Under Kantian ethics, it is considered unethical to use individuals merely as a means to an end, without respecting their intrinsic value and autonomy. This principle raises significant concerns about the business models of companies that seek to maximize user engagement, particularly in the context of social media.

In the case of social media platforms, the business model has evolved to focus on keeping users engaged for as long as possible. This model often relies on collecting user data to sell to advertisers, effectively using the users as a means to generate revenue. While the initial intent of social media might have been to connect people and provide entertainment, the current practices prioritize engagement metrics and ad revenue over genuine user well-being.

Social media companies have developed engagement algorithms designed to keep users hooked, sometimes by promoting content that generates outrage or polarization [2]. This approach, while effective at increasing user time on the platform, can exploit users' attention and emotions, aligning with the Kantian critique that these companies are using individuals merely as a means to achieve their financial goals. The shift from a focus on meaningful connections to prioritizing engagement and revenue highlights the ethical concerns surrounding the manipulation of users for profit.

B. Rules Utilitarianism

From the perspective of rule utilitarianism, the ethical evaluation of a company's product being financially incentivized to keep users engaged indefinitely would involve assessing whether such a rule would generally lead to the greatest overall happiness and minimize suffering.

If the rule allows for products that are potentially infinitely addictive, it could lead to significant negative consequences. The suffering caused by addiction—whether from social media, addictive drugs, or other products—can be profound. Addicts may experience detrimental effects on their mental health, relationships, and overall well-being, all of which contribute to considerable harm and suffering.

Even if companies stand to make substantial profits from such products, these financial gains do not necessarily compensate for the widespread harm inflicted on individuals. Rule utilitarianism would likely find that a rule permitting products to be designed solely to maximize user engagement indefinitely would lead to more overall suffering than happiness. The harm caused by addiction, which often includes both personal distress and broader societal impacts, would outweigh any benefits derived from the profits of such companies.

V. CONCLUSION

A product whose sole financial incentive is to ensure users continue using it is inherently unethical. Such products are often designed with addictive features to keep users engaged for as long as possible. This approach is problematic because it prioritizes profit over the well-being of users. By deliberately making a product addictive, companies are using their users merely as a means to generate revenue, rather than considering their welfare. Addiction can lead to significant harm not only to the users themselves but also to those around them, making this business model both exploitative and damaging.

Casinos are fundamentally driven by the incentive to keep their customers gambling for as long as possible, which renders their business model unethical. Most governments around the world acknowledge this issue and therefore impose stringent regulations on gambling. While gambling can be seen as a form of entertainment, its potential for addiction leads to significant concerns. Consequently, many countries permit consenting adults to gamble, recognizing the need for personal choice while striving to mitigate the associated risks.

Certain video games are heavily incentivized to incorporate addictive design elements to promote the sale of microtransactions, which can themselves be addictive. This monetization model is considered unethical because it prioritizes profit over the well-being of players. Given the similarities between these practices and those in gambling, video games that rely on such addictive microtransactions should be subject to government regulation similar to that imposed on gambling to protect consumers.

Social media platforms, which primarily generate revenue from the data collected while users engage with their sites, can be deemed unethical due to their financial incentives to incorporate addictive design elements. However, this does not necessarily mean that social media should be dismantled, as it has also provided significant benefits by connecting people worldwide. Instead of eliminating these platforms, we should explore alternative revenue sources for social media companies to reduce their reliance on addictive designs. Additionally, governments may need to approach social media regulation differently. Drawing on the examples of tobacco, alcohol, and gambling, education should be a key strategy. Raising awareness and educating users about the risks associated with addictive design in social media can help mitigate potential harms.

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